Research output of three CSIR chemistry labs, 2016–2026

CSIR-NCL (Pune) · CSIR-IICT (Hyderabad) · CSIR-CECRI (Karaikudi) · OpenAlex, collected 13 Oct 2026 (2026 partial)

Institute	Papers	Journal	Conference	Author rows	Citations	Q1 (SJR)	Roster scientists	With ≥1 paper
CSIR-IICT	6,123	6,045	78	20,073	125,172	55.0%	168	129
CSIR-NCL	5,316	5,269	47	14,471	133,859	67.3%	166	151
CSIR-CECRI	2,870	2,816	54	7,445	93,099	73.1%	51	26
All three (distinct)	14,208	14,029	179	41,989	350,015	63.2%	385	306

How the data was built

- OpenAlex works of the three ROR-verified institutes; articles, reviews, letters and conference papers
- Faculty = scientific cadre on official staff pages, matched to OpenAlex authors in the corpus
- Quartile = SCImago (SJR) best quartile of the publication year (2025 values for 2026)

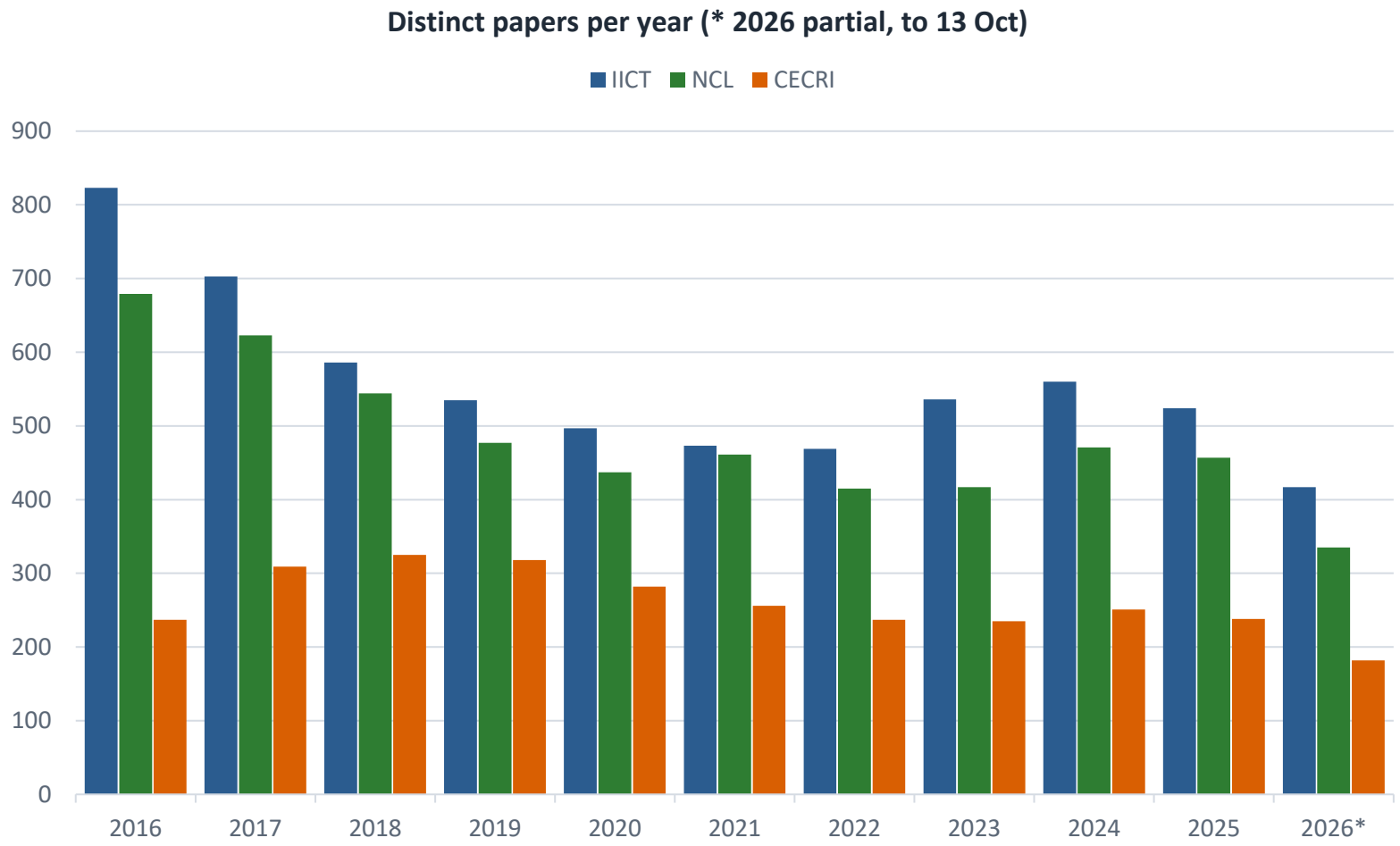
Limits

- OpenAlex author and affiliation matching is imperfect; papers may be missing or mis-attributed
- h-index uses 2016–2026 corpus papers only; students and project staff are not faculty
- SJR may differ from JCR; none of the 179 conference papers is CORE-ranked

Interactive dashboard: https://dpcnsite.vercel.app/

Publication trend

14,208 distinct papers in 2016–2026; IICT 43% of them, NCL 37%, CECRI 20%



Totals and growth

IICT: 6,123 papers;
CAGR 2016→2025 −4.9%; peak year 2016

NCL: 5,316 papers;
CAGR 2016→2025 −4.3%; peak year 2016

CECRI: 2,870 papers;
CAGR 2016→2025 +0.1%; peak year 2018

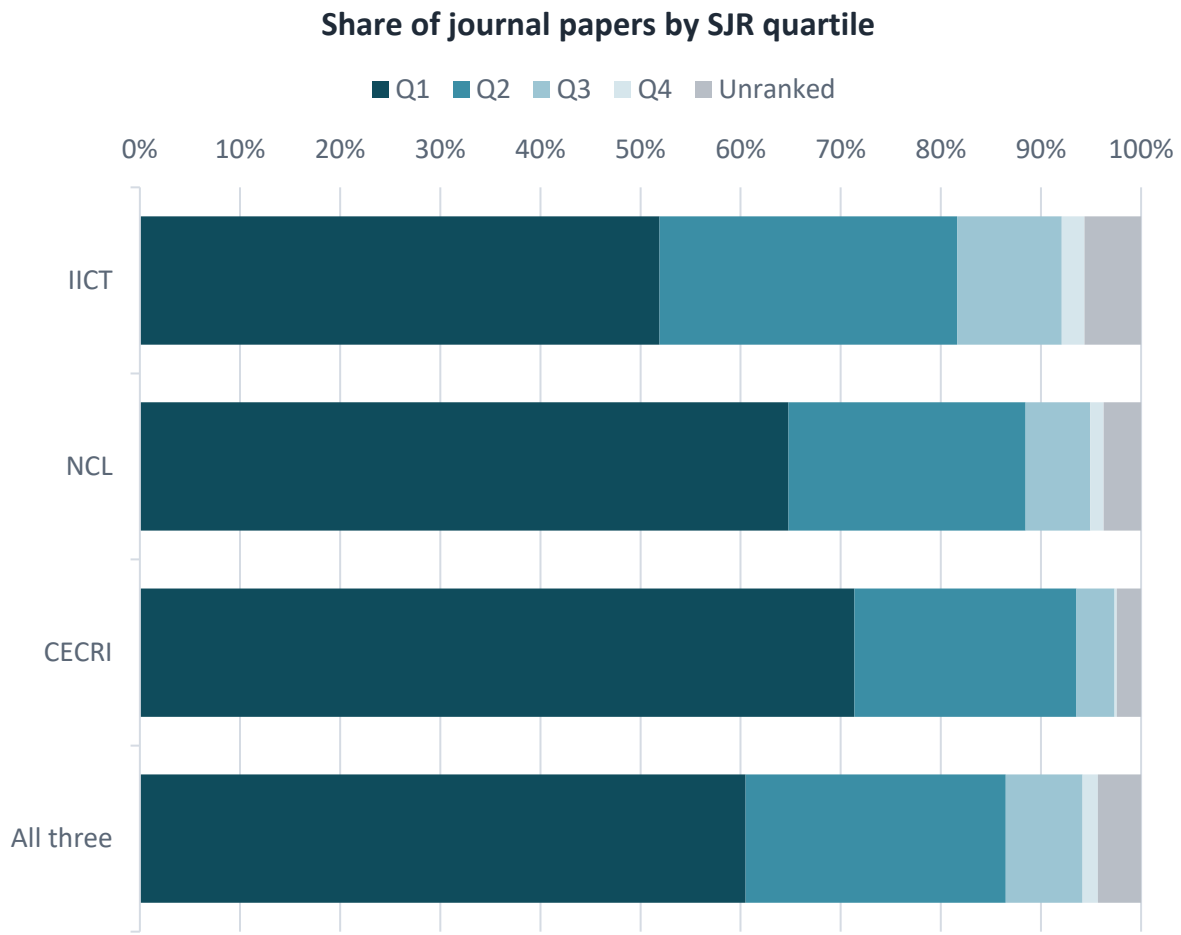
Observations

- Combined output went from 1,727 papers in 2016 to 1,208 in 2025
- Growth rates differ in sign across the three labs
- 2026 covers papers to 13 Oct only (924 so far) and is left out of CAGR

Source: stats/tables/publication_trend.csv · Papers counted once per institute; combined counts distinct papers

Journal quartiles and citation impact ratio

63.2% of ranked journal papers are in SJR-Q1 journals; Q1 by impact ratio (IR > 0.6, top 25% of papers) ranges from 21.5% (IICT) to 33.5% (CECRI)



Q1 by journal vs Q1 by impact ratio (papers, % of papers)

Institute	Q1 by journal (SJR)	Q1 by IR (IR > 5)	Q1 by IR (IR > 10)	Q1 by IR (IR > 0.6)	Institute IR
IICT	55.0%	14 (0.24%)	4 (0.07%)	21.5%	0.41
NCL	67.3%	42 (0.82%)	12 (0.23%)	23.8%	0.48
CECRI	73.1%	17 (0.60%)	2 (0.07%)	33.5%	0.60
All three	63.2%	73 (0.53%)	18 (0.13%)	24.8%	0.47

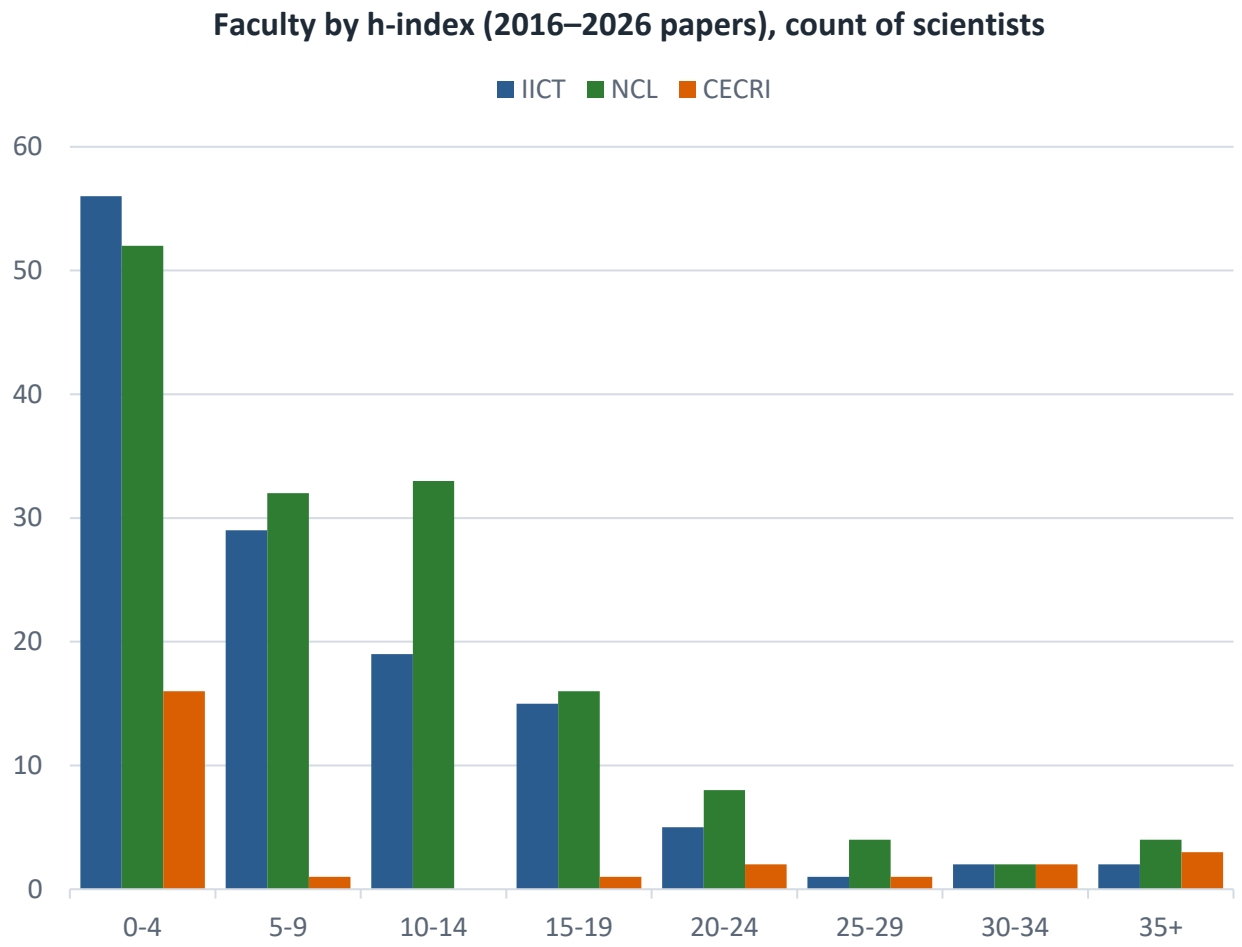
Impact ratio (IR) = citations received ÷ citations made (references). A paper is IR-Q1 when IR exceeds the threshold; the institute IR is all citations received ÷ all references made.

- Threshold IR > 0.6 is the 75th percentile of IR over all papers, so IR-Q1 is the top 25% of papers — the same share a quartile covers (median paper IR 0.26).
- The two definitions disagree: only 32.4% of papers in SJR-Q1 journals are top-25% by IR. SJR ranks the journal; IR ranks the paper.
- Older papers have had longer to be cited; 451 papers with no references in OpenAlex are excluded.
- IR > 5: 73 papers, of which 57 have ≥10 references; the rest have short OpenAlex reference lists that overstate IR (list: impact_ratio_high_papers.csv).

Source: stats/tables/quartile_distribution.csv, impact_ratio_summary.csv, impact_ratio_vs_sjr.csv · SJR best quartile (Scopus-based); IR from OpenAlex citations (13 Oct) and references

Faculty h-index distribution

Median h-index of the 306 roster scientists with papers is 7; CECRI 2.5, NCL 8, IICT 6



Institute	Faculty	Median	Mean	P75	Max
IICT	129	6	8.1	12	54
NCL	151	8	9.8	14	43
CECRI	26	2.5	11.9	23.5	59
All	306	7	9.2	13.75	59

Highest h-index (2016–2026 window)

Top faculty by h-index	Lab	Papers	Citations	h
Dr Subrata Kundu	CECRI	258	15,782	59
Dr. S. Venkata Mohan	IICT	226	11,334	54
Dr. Sreekumar Kurungot	NCL	177	8,794	43
Dr. Subashchandrabose Chi…	NCL	57	3,617	41
Dr. Ujjwal Pal	IICT	148	4,758	40
Dr. Vinod C Prabhakaran	NCL	149	4,737	39

Source: stats/tables/h_index_summary.csv, h_index_distribution.csv, faculty_performance.csv · h-index on 2016–2026 corpus papers

Top 5 journals per institute

Most frequent venues: IICT – ChemistrySelect (239); NCL – ChemistrySelect (140); CECRI – Electrochimica Acta (77)

CSIR-IICT 6,123 papers

#	Journal	Papers	Share	Q
1	ChemistrySelect	239	4.0%	Q2
2	Organic & Biomolecular Chemistry	144	2.4%	Q1
3	Tetrahedron Letters	128	2.1%	Q2
4	ChemInform	122	2.0%	—
5	The Journal of Organic Chemistry	120	2.0%	Q1

CSIR-NCL 5,316 papers

#	Journal	Papers	Share	Q
1	ChemistrySelect	140	2.7%	Q2
2	Chemical Communications	116	2.2%	Q1
3	The Journal of Organic Chemistry	106	2.0%	Q1
4	ACS Omega	97	1.8%	Q1
5	Organic Letters	94	1.8%	Q1

CSIR-CECRI 2,870 papers

#	Journal	Papers	Share	Q
1	Electrochimica Acta	77	2.7%	Q1
2	Journal of Materials Chemistry A	68	2.4%	Q1
3	ChemistrySelect	66	2.3%	Q2
4	New Journal of Chemistry	64	2.3%	Q1
5	International Journal of Hydrogen…	59	2.1%	Q1

All three combined: ChemistrySelect (442); New Journal of Chemistry (269); RSC Advances (241); Chemical Communications (237); The Journal of Organic Chemistry (225)

Source: stats/tables/top_journals.csv · ranked by distinct papers; share = % of the institute's papers; Q = SJR quartile

Top 10 most-cited papers (all three institutes)

350,015 citations in total; mean 24.6 and median 11 per paper; the top paper has 2,493

#	Title	Journal	Year	Lab	Citations	DOI
1	Recent Trends and Perspectives in Electrochemical Water Splitting with an Emphasis on Sulfi…	ACS Catalysis	2016	CECRI	2,493	10.1021/acscatal.6b02479
2	Cancer chemotherapy and beyond: Current status, drug candidates, associated risks and progr…	Genes & Diseases	2022	IICT	1,826	10.1016/j.gendis.2022.02.007
3	Precision and correctness in the evaluation of electrocatalytic water splitting: revisiting…	Energy & Environmental Science	2018	CECRI	1,623	10.1039/c7ee03457a
4	Covalent Organic Frameworks: Chemistry beyond the Structure	Journal of the American Chemi…	2018	NCL	1,574	10.1021/jacs.8b10334
5	Phytochemicals in Cancer Treatment: From Preclinical Studies to Clinical Practice	Frontiers in Pharmacology	2020	NCL	1,085	10.3389/fphar.2019.01614
6	Selective Molecular Separation by Interfacially Crystallized Covalent Organic Framework Thi…	Journal of the American Chemi…	2017	NCL	1,060	10.1021/jacs.7b06640
7	Cooperativity in Noncovalent Interactions	Chemical Reviews	2016	IICT	932	10.1021/cr500344e
8	“The Fe Effect”: A review unveiling the critical roles of Fe in enhancing OER activity of N…	Nano Energy	2020	CECRI	931	10.1016/j.nanoen.2020.105514
9	Functional Naphthalene Diimides: Synthesis, Properties, and Applications	Chemical Reviews	2016	IICT	927	10.1021/acs.chemrev.6b00160
10	Polylactic acid: synthesis and biomedical applications	Journal of Applied Microbiolo…	2019	NCL	873	10.1111/jam.14290

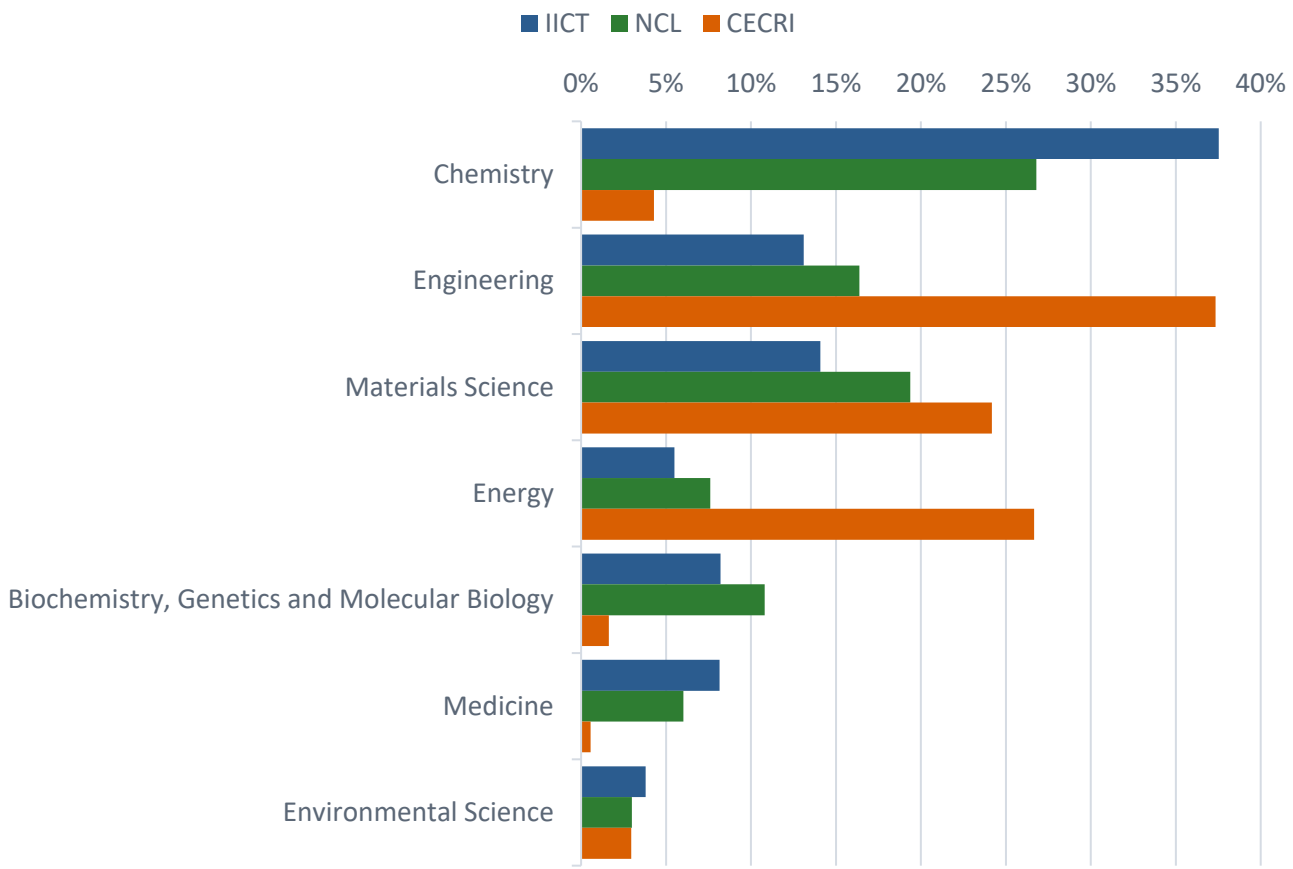
IICT: 125,172 citations, mean 20.4, uncited 11.9% · NCL: 133,859 citations, mean 25.2, uncited 10.6% · CECRI: 93,099 citations, mean 32.4, uncited 6.5%

Source: stats/tables/top_cited_papers.csv, citation_summary.csv · OpenAlex cited_by_count on 13 Oct 2026; older papers have had longer to accrue citations

Research areas and departments

Largest field: IICT – Chemistry (38%), NCL – Chemistry (27%), CECRI – Engineering (37%)

Share of each institute's papers by OpenAlex field



Top 2 departments per institute by papers

Department	Lab	Faculty	Papers	Q1
Polymers & Functional Materials	IICT	15	492	68%
Applied Biology	IICT	16	477	56%
Physical and Materials Chemistry	NCL	26	1,171	74%
Organic Chemistry	NCL	37	745	60%
Electrochemical Process Engineering	CECRI	6	348	90%
Electrodics and Electrocatalysis	CECRI	2	142	75%

Department = roster division of each faculty member; a paper counts in every department of its faculty authors. Papers by non-roster authors only are not assigned to a department.

Source: stats/tables/research_areas_field.csv, department_performance.csv · field = OpenAlex primary topic → field

Citation network

Faculty level: 246 of 306 scientists linked by 1,534 'A cites B' links, 294 across labs. Paper level: 27,099 links among 10,931 papers

1,534
citation links between the 306 faculty scientists

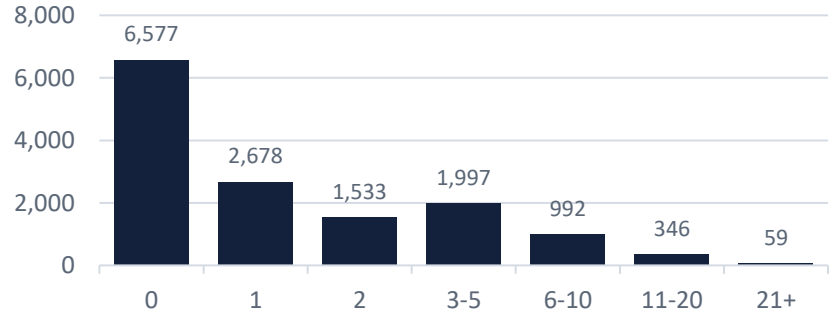
10,931
papers in the paper-level graph (nodes are papers, not people)

1,639
paper citation links between the three labs

1.02
in / out citation ratio with other Indian institutions

#	PageRank top 10	Lab	Year	In-deg	PR
1	Recent Trends and Perspectives in Electrochemical Water…	CECRI	2016	129	0.0030
2	Pt Nanoparticle Anchored Molecular Self-Assemblies of D…	CECRI	2016	41	0.0029
3	Unprotected and interconnected Ru 0 nano-chain networks…	CECRI	2016	30	0.0027
4	Functional Naphthalene Diimides: Synthesis, Properties,…	IICT	2016	52	0.0020
5	Precision and correctness in the evaluation of electroc…	CECRI	2018	110	0.0018
6	Waste biorefinery models towards sustainable circular b…	IICT	2016	71	0.0017
7	Regulation of acidogenic metabolism towards enhanced sh…	IICT	2016	40	0.0016
8	Pharmaceutical cocrystals: walking the talk	NCL	2016	35	0.0016
9	Facile and scalable route to sheets-on-sheet mesoporous…	CECRI	2016	18	0.0015
10	Recent advances in N-heterocyclic carbene (NHC)-catalys…	NCL	2016	27	0.0014

In-degree distribution (corpus papers)



Citation flow with other Indian institutions

Lab	Cited by Indian	Cites Indian	In/out	Citing insts
IICT	32,818	38,814	0.85	1,816
NCL	32,264	29,926	1.08	1,683
CECRI	23,355	18,172	1.28	1,310

Source: stats/tables/network_summary.csv, network_citation_*.csv, citation_flow_summary.csv · in-degree = citations from other corpus papers; flows count links to DOI-bearing papers

Authorship network

306 faculty scientists: 553 co-author pairs, only 16 across labs; 14 communities. All 7,853 lab authors (students and staff included): 35,392 pairs

306
faculty scientists (nodes); 234 have a faculty co-author

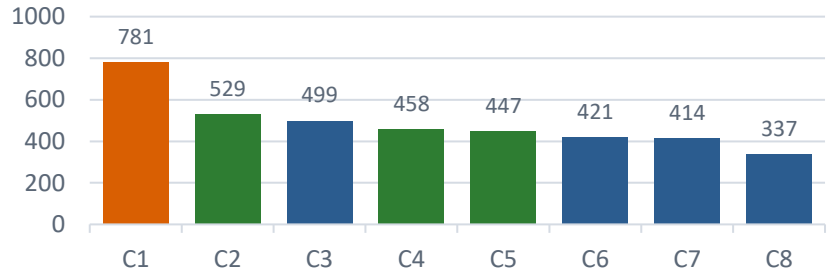
553
faculty co-author pairs; 16 join two labs

7,853
all lab authors: 283 faculty + 7,570 students, postdocs, staff

134
communities in the all-author graph (modularity 0.82)

#	Most-connected faculty	Lab	Fac. co-auth.	Shared papers	Betweenness	Btw rank
1	Dr. Rajesh G. Gonnade	NCL	38	196	0.095	#5
2	Dr. Kumar Vanka	NCL	31	152	0.116	#3
3	Dr. Jagadeesh Babu Nanubolu	IICT	22	79	0.142	#1
4	Dr. Sunil Misra	IICT	22	52	0.114	#4
5	Dr. Dhiman Sarkar	NCL	21	29	0.122	#2
6	Dr. Sailaja Krishnamurty	NCL	20	78	0.028	—
7	Dr. Arun Torris	NCL	19	75	0.018	—
8	Dr. Andugulapati Sai Balagi	IICT	19	44	0.053	#7
9	Dr. Sreekumar Kurungot	NCL	18	79	0.018	—
10	Dr. R. Nandini Devi	NCL	15	35	0.012	—

Largest communities, all lab authors (colour = main lab)



Faculty brokers by betweenness

Dr. Jagadeesh Babu Nanubolu IICT · 0.142 · 22 faculty co-authors

Dr. Dhiman Sarkar NCL · 0.122 · 21 faculty co-authors

Dr. Kumar Vanka NCL · 0.116 · 31 faculty co-authors

Dr. Sunil Misra IICT · 0.114 · 22 faculty co-authors

Top Indian partners: Academy of Scientific and Innovative Research (635); Savitribai Phule Pune University (372); Council of Scientific and Industrial Research (334)

Source: stats/tables/network_summary.csv, network_authorship_*.csv, indian_partner_institutions.csv · interactive graphs: https://dpcnsite.vercel.app/

Ten non-obvious findings

Each finding answers one question; charts and method notes are in the dashboard's Insights tab

#	Finding	Result
1	Friendship paradox	93.2% of researchers have fewer co-authors than their co-authors do. The average researcher has 9.1 co-authors, but the average co-author has 30.8.
2	Rich club	The top 1% most-connected researchers show no rich-club effect and are less interconnected than chance (ρ_norm = 0.72). 40 of 79 are CSIR faculty.
3	Network fragility	Removing the top 908 bridge researchers (11.7% of the network) halves the cluster; random removal needs 3,367. Removing 541 of them separates the three labs' faculty into different components.
4	Concentration (Price's law)	√N = 18 scientists produce 36.6% of the cluster's papers (Price's law predicts 50%). The top 10% produce 48.2%; Gini = 0.61 (papers), 0.74 (citations).
5	Journal quartile vs impact	Q1 is a strong filter but not a guarantee: 88.7% of the cluster's top-10% most-cited papers appeared in Q1 journals, yet only 14.0% of Q1 papers reach the top 10% (vs 3.5% of Q2–Q3 papers).
6	Local vs global influence	In-cluster influence and global citations correlate only moderately (Spearman ρ = 0.40). 45 papers are 'hidden gems': top-5% internal influence, below-median global citations.
7	Sleeping beauties	2 of the 7 strongest candidates (top 1% of 601 papers by Ke's B) show sustained delayed recognition; 3 are short citation spikes and 2 peaked in 2025, too recent to tell. The highest-B sustained case, ranked 4 of 7 by B, 'Funnel shaped molecules containing benzo/pyrido[1,2,5]thiadiazole f…' (IICT, 2016), drew 1.4 citations/yr for 5 years, then 4.7/yr from 2021.
8	COVID shock	Output fell 8.6% from 2019 to 2020 (1,320 → 1,207 papers) and was 10.2% below 2019 in 2021, but it had already been falling 8.6% a year since 2016: against that trend 2020 was 2.2% above and 2021 13.9% above, so there is no COVID-specific dip. COVID-related work peaked at 5.5% of CSIR-IICT's papers in 2021; the topic mix shifted about as much as in a normal period (JSD ratio 1.04).
9	Brain circulation	17.0% of classifiable CSIR scientists are returnees (NCL 17.7% (n=141), IICT 17.0% (n=106), CECRI 11.1% (n=18)); their abroad spell was most often in the US (25 of 45). Their papers' citations do not differ significantly from home-grown peers of similar career age (+9%, 95% CI -19% to +47%); 20.5% of returnees' papers vs 16.7% of home-grown scientists' papers have a foreign co-author.
10	Succession risk	In CECRI, 46.3% of recent papers (36.4–51.1% for age cut-offs 32–28; 48.6% of their citations) have only roster authors likely to retire within ~5 years (IICT 20.9%, NCL 11.7%). Highest exposure: Electrochemical Process Engineering, CECRI (74.5% of 192 papers).

Source: stats/tables/insights/*.json (generated by src/stats/insights) · charts: https://dpcnsite.vercel.app/